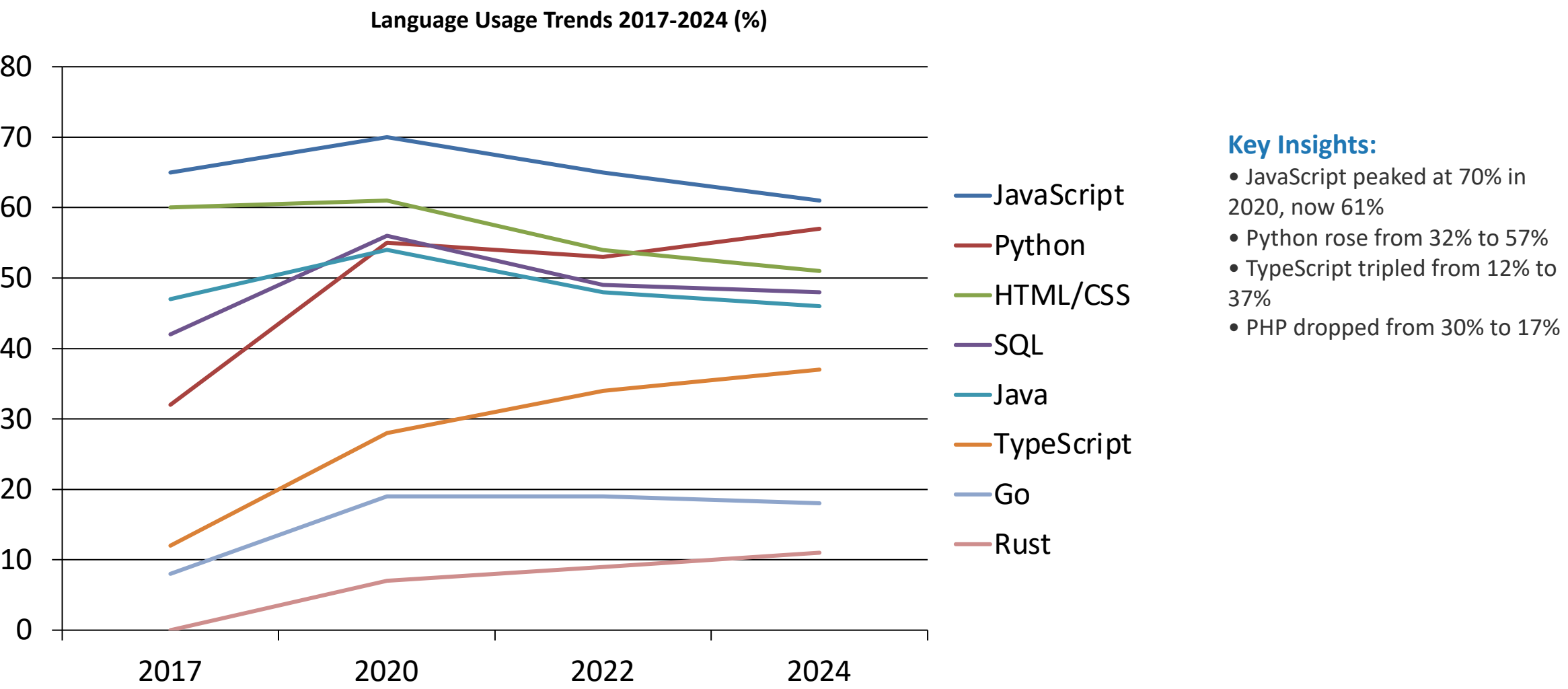


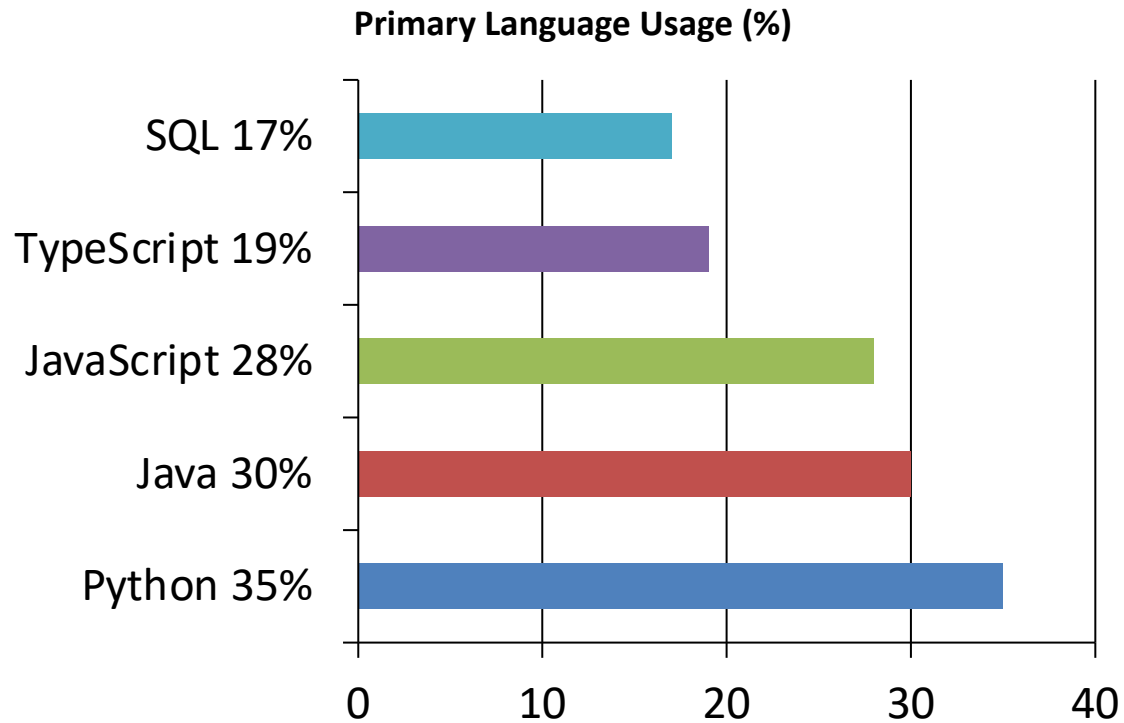
State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

JetBrains Survey of 23,262 Developers Worldwide

Python Closes the Gap: 8 Years of Language Usage Shifts



Python Is Now the #1 Primary Language at 35%



Adoption Intent (Plan to Adopt)

Python: 6%

TypeScript: 5%

Java: 4%

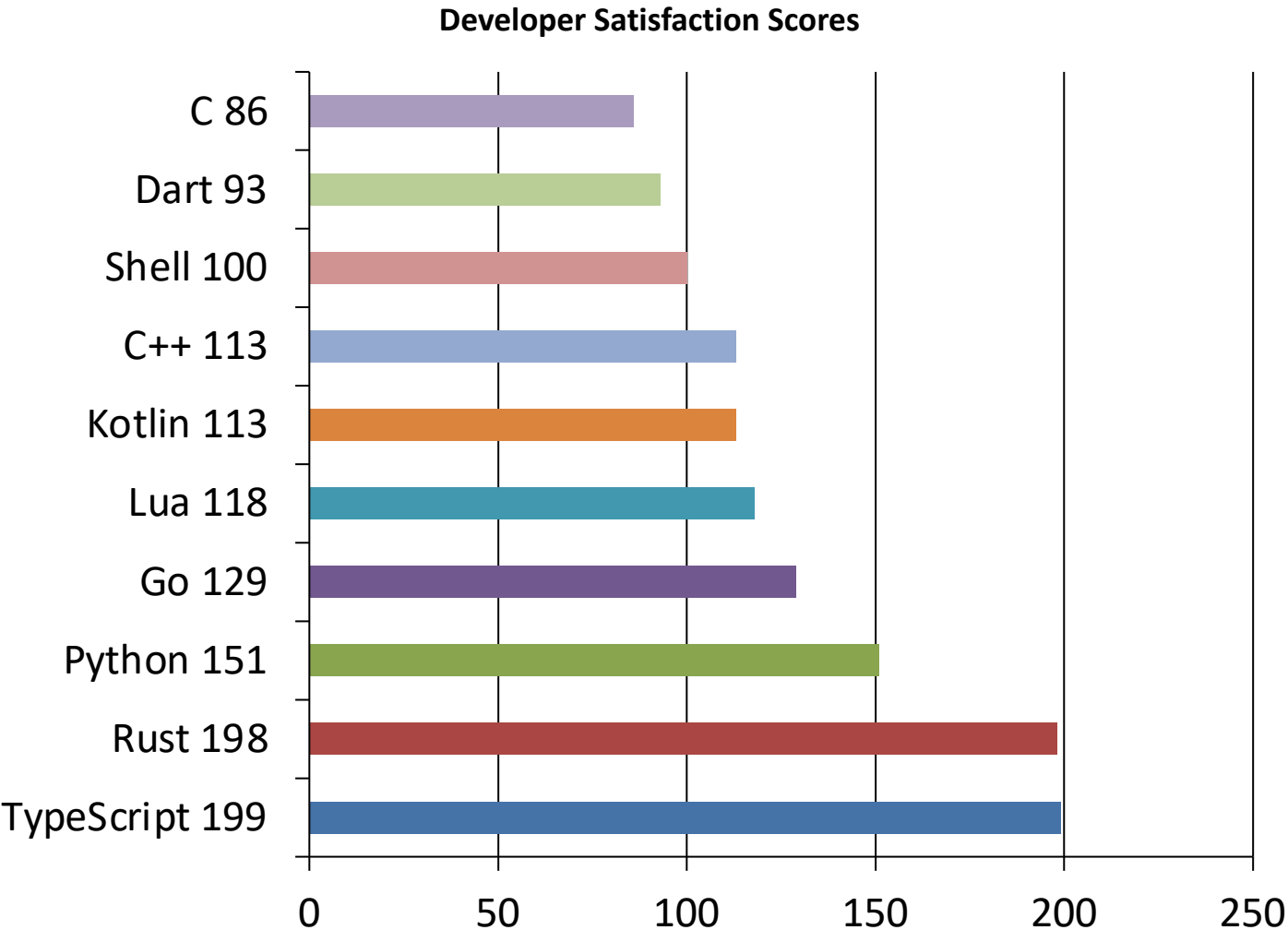
SQL: 4%

Go: 3%

Rust: 3%

Insight: Ecosystem lock-in keeps developers loyal. Adoption intent stays below 6% across all languages.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs



Key Insight

Type-safe, modern-tooling languages dominate the top spots.

Why TypeScript & Rust?

- Type safety and compile-time checks
- Modern developer experience
- Expressive type systems
- Strong tooling ecosystems

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Top Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
Data Science / Analytics	Python	28%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%

Key Insight

Language choice is driven by domain, not general preference.

JavaScript + TypeScript own UI development (58% + 62%)
Python dominates AI/ML & data (31%, 28%, 17%)
Java leads backend services (49%, 55%)

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML teams

Python is the undisputed leader in artificial intelligence, machine learning, and data science — driven by the AI boom.

2

TypeScript is the future of frontend

With 62% satisfaction and 37% usage growth, TypeScript owns the UI development space and full-stack roles.

3

Rust & Go are growth plays

Both languages show steady growth and high developer satisfaction — ideal for infrastructure and systems teams.

4

PHP & Ruby pools are shrinking

PHP dropped from 30% to 17%; Ruby fell from 10% to 4%. Factor shrinking talent pools into long-term decisions.

5

Satisfaction = type safety

TypeScript (199) and Rust (198) lead satisfaction. Modern tooling and type systems drive retention.